

PhD Qualification Exam in Applied Mathematics

Mock Examination No. 1

Total: 100 points

Problems 1–5 are required. Solve exactly one of Problems 6 and 7. If both are attempted, only the higher score will be counted.

1. (15 points) Let $I(f) = \int_{-1}^1 (1+x)f(x) dx$. A two-node quadrature rule $Q(f) = w_-f(x_-) + w_+f(x_+)$ is required to have the largest possible degree of exactness.

- Construct the monic quadratic polynomial orthogonal to 1 and x in the weighted inner product induced by I , and hence determine x_- and x_+ .
- Determine the two weights and prove that they are positive.
- State the degree of exactness. If $f \in C^4[-1, 1]$, derive the error formula and determine its sign when $f^{(4)} \geq 0$.

2. (15 points) For a parameter $\beta > 0$, consider $f(x) = xe^x - \beta$.

- Prove that f has a unique positive zero α , and locate it in a finite interval depending only on β .
- The equation may be written as $x = \beta e^{-x}$. Determine exactly for which β the corresponding fixed-point iteration is locally contractive at α , and find its asymptotic error factor.
- Prove that Newton's method started at $x_0 = \beta$ is well defined, decreases monotonically to α for every $\beta > 0$, and is eventually quadratically convergent.

3. (20 points) Let $A \in \mathbb{R}^{m \times n}$ have full column rank, and let $b \in \mathbb{R}^m$. For $\lambda > 0$ define

$$x_\lambda = \arg \min_x (\|Ax - b\|_2^2 + \lambda \|x\|_2^2).$$

- Derive the normal equations and prove uniqueness.
- Using a thin singular value decomposition $A = U\Sigma V^T$, derive the filter-factor representation of x_λ and the limit as $\lambda \downarrow 0$.
- Prove that $\|x_\lambda\|_2 \leq \|x_*\|_2$ and $\|Ax_\lambda - b\|_2 \geq \|Ax_* - b\|_2$, where x_* is the unregularized least-squares solution.
- Compute $\kappa_2(A^T A + \lambda I)$ and explain how an SVD can be reused efficiently when solutions are required for many values of λ .

4. (15 points) On a periodic grid, consider $u_t + au_x = \nu u_{xx}$ with $\nu > 0$ and the IMEX scheme

$$\frac{U_j^{n+1} - U_j^n}{\tau} + a \frac{U_{j+1}^n - U_{j-1}^n}{2h} = \nu \frac{U_{j+1}^{n+1} - 2U_j^{n+1} + U_{j-1}^{n+1}}{h^2}.$$

Determine its local truncation error. Using Fourier analysis, find a necessary and sufficient condition for ℓ^2 stability in terms of a , ν , and τ .

5. (15 points) Let $0 = x_0 < x_1 < \dots < x_M = 1$, $h_{i+1/2} = x_{i+1} - x_i$, and $\bar{h}_i = (h_{i-1/2} + h_{i+1/2})/2$. For $a_{i+1/2} > 0$ and $c_i \geq c_0 > 0$, define

$$(L_h U)_i = -\frac{1}{\bar{h}_i} \left[a_{i+1/2} \frac{U_{i+1} - U_i}{h_{i+1/2}} - a_{i-1/2} \frac{U_i - U_{i-1}}{h_{i-1/2}} \right] + c_i U_i.$$

Prove a discrete maximum principle for L_h . Deduce that the solution of $(L_h U)_i = f_i$ with prescribed endpoint values satisfies

$$\|U\|_\infty \leq \max \left\{ |U_0|, |U_M|, \frac{\|f\|_\infty}{c_0} \right\}.$$

6. (20 points; choose one of Problems 6–7) Let $a \in C^4[-1, 1]$. For

$$I(\varepsilon) = \int_{-1}^1 a(x) e^{ix^2/\varepsilon} dx, \quad \varepsilon \downarrow 0.$$

derive the stationary-point contribution through its first correction and the leading endpoint contribution. Order all terms by size, justify the remainder, and state what becomes the leading term if $a(0) = 0$ but $a(1) + a(-1) \neq 0$.

7. (20 points; choose one of Problems 6–7) For a fixed integer $N \geq 1$, consider the finite-horizon control problem

$$x_{k+1} = x_k + u_k, \quad k = 0, \dots, N-1, \quad J = \sum_{k=0}^{N-1} u_k^2 + x_N^2,$$

where x_0 is given. Write the Bellman recursion, derive the Riccati recursion for the quadratic value functions, and obtain closed forms for the optimal feedback controls, state trajectory, and optimal value. State the arithmetic complexity of the backward dynamic-programming pass.